

Experimental Load Emulation for Multi-Evaporator Air Conditioning and Refrigeration Systems

Justin KOELN*, Megan KANIA, Neera JAIN, Andrew ALLEYNE

Department of Mechanical Science and Engineering,
University of Illinois at Urbana-Champaign,
1206 West Green Street, Urbana, IL 61801, USA
(Phone: +1-217-244-6556, Fax: +1-217-244-6534, E-mail: koeln2@illinois.edu)

ABSTRACT

This paper presents the development of a multi-evaporator load emulation unit (ME-LEU) for air conditioning and refrigeration (AC&R) systems. Through a combination of on-line real-time modeling and hardware actuation, LEUs are able to mimic the dynamics of a variety of cooling loads. A virtual container model simulates the thermal dynamics of the cooled space and determines the appropriate return air temperature to the evaporator for a given application. The LEU uses an actuated slide to mix warm return air with colder supply air to track the desired return air temperature from the model. Depending on the physical design of the LEU, control of the return air temperature may become difficult due to non-monotonic and nonlinear dynamics. The presented LEU design achieves monotonic and nearly linear dynamics between the actuation mechanism and the return air temperature, allowing for a simple yet robust controller to be designed for the system. The experimental implementation of the ME-LEU on a dual evaporator AC&R system permits independent loading of each evaporator, thereby increasing the emulation capabilities to a larger set of real world systems.

1. INTRODUCTION

Multi-evaporator air conditioning and refrigeration (ME-AC&R) systems are widely used in buildings, supermarkets, and other applications where it is preferable to drive a single compressor and use a network of evaporators to cool distributed spaces with varying load profiles and disturbances (Owen 2008). The coupled nature of these distributed spaces makes them more challenging to model dynamically, thereby making it more difficult to implement effective temperature control in the spaces. The typical temperature controller development process includes significant test facility time. This paper describes the development and capabilities of an alternative: a multi-evaporator load emulation unit (ME-LEU). Through a combination of real-time modeling and hardware actuation, an LEU is able to simulate the dynamics of various loads without the footprint and cost of a large-scale calorimeter cell (Otten *et al.* 2010). Therefore, a variety of real world cooling loads and disturbances can be emulated on an experimental ME-AC&R system in order to validate dynamic models as well as to develop and test more efficient and higher performance controllers.

The LEU provides a similar function for an AC&R system as a dynamometer does for an engine. Dynamometers are used to apply a simulated load to an engine in a laboratory setting (Martyr and Plint 2012). The load on an engine is the product of the angular velocity and torque of the engine. The simulated angular velocity profile comes from a drive profile such as the EPA Urban Dynamometer Driving Schedule (UDDS) (Kruse and Huls 1973), and the simulated torque profile comes from a simulation of various forces acting on the vehicle such as wind resistance, rolling friction, etc. (Babbitt and Moskwa 1999). The dynamometer then applies this realistic load to the engine so that engine performance and control strategies can be studied under different conditions that the engine might encounter. Similarly, in the ME-LEU, a virtual container model (VCM) calculates the thermal dynamics of a cooled space and determines the thermal load to be placed on the AC&R system. In this case, the thermal load is a product of the mass flow and temperature of the cooling medium entering the evaporator (air for the current study). The fan on the evaporator provides the air mass flow rate and the LEU provides the ability to deliver a prescribed temperature to the evaporator. The VCM uses parameters of the cooled space, such as volume and wall materials, and thermal loads, such as ambient temperature, wind, and solar radiation, to determine an average temperature for the interior of the cooled space. This return air temperature is then provided to the inlet of the evaporator via the LEU, which mixes warm ambient air with cool supply air from the outlet of the evaporator. In this way, the

combination of the VCM and LEU is able to apply a wide range of realistic loads to the AC&R system in order to study the performance of the system and develop improved control strategies.

To provide the simulated load from the VCM, it is necessary for the LEU to be able to track a wide range of return air temperature dynamics. In Otten *et al.* (2010), it was shown that the physical design of the LEU has a large impact on the controllability of evaporator return air temperature. In particular, certain design choices resulted in air flow patterns such that the relationship between the control input, which governs the mixing rate, and return air temperature was not only nonlinear but also non-monotonic. Moreover, rapid oscillations in actuator position and poor load tracking resulted due to the variable sensitivity between actuator position and return air temperature. The load controller performance was sub-optimal, thereby limiting the capability of the LEU and resulting in a need to redesign the LEU.

In this paper we present a multi-evaporator load emulation unit (ME-LEU) design which addresses the deficiencies of the original single LEU and also expands its capabilities. The ME-LEU features streamlined flow paths designed to reduce turbulence and vorticity in the evaporator return and supply airflow, thereby enabling improved reference tracking of a wider range of thermal loads. The capabilities of the ME-LEU are demonstrated on a dual-evaporator AC&R experimental test system. Case studies are presented in which independent and coupled loads and disturbances are applied to each evaporator. With the ability to emulate multiple loads, a larger set of real world systems may be studied in an effort to develop more effective control methods for improved efficiency and performance.

This paper is organized as follows. Section 2 describes the design of the ME-LEU and characterization of its dynamic behavior. Section 3 details the VCM and how this model interfaces with the physical test system. The controller design for each LEU in the ME-LEU is presented in Section 4. In Section 5, the performance and capabilities of the ME-LEU are presented through various experimental case studies. Finally, the main points of the paper are summarized in the Conclusion.

2. SYSTEM DESIGN AND DYNAMICAL ANALYSIS

Figure 1 shows a model of the ME-LEU. The LEUs can be connected with a duct containing an interchangeable wall, allowing for variable thermal resistance. This duct provides the capability to simulate the thermal coupling between cooled spaces seen in many real world systems. However, the current study does not use this duct, allowing the dynamics of each LEU to be decoupled; the thermal coupling provided by this duct is left for future work. Since the two LEUs that comprise the ME-LEU are identical, the same plant model and controller can be used for both.

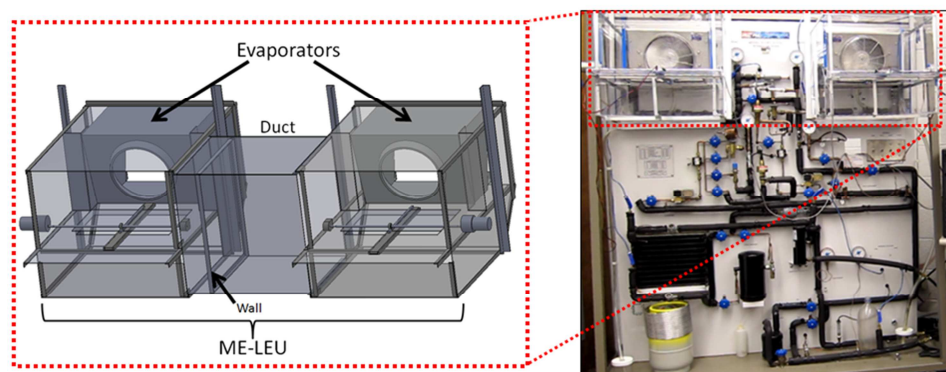


Figure 1: Model of ME-LEU with thermal coupling duct.

To determine the plant model, a single LEU of the new design (shown in Figure 2) was studied. The return temperature is controlled using a linear slide which adjusts the mixture of warm ambient air and cool supply air. Figures 3a and 3b show the control volume schematics of region *a*, the top section of the LEU, and region *b*, the lower section of the LEU.

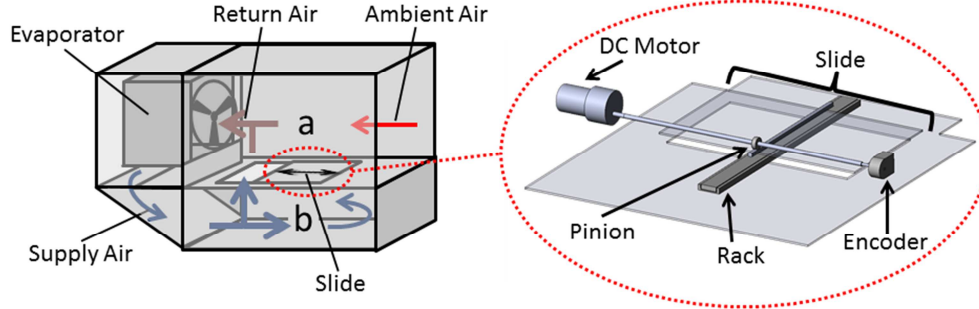


Figure 2: LEU design with adjustable slide and airflow paths.

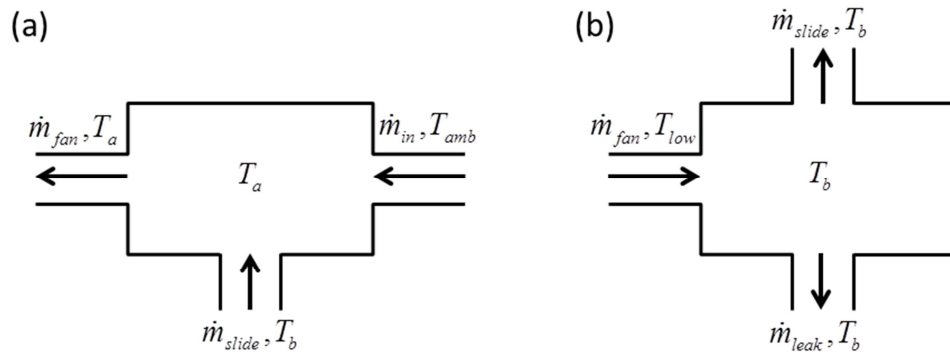


Figure 3: Control volume schematics for regions *a* and *b* of the LEU.

In region *b*, cool supply air from the exit of the evaporator enters the control volume with temperature T_{low} and mass flow rate $\dot{m}_{fan}$, and air leaves the control volume through 1) the slide with temperature T_b and mass flow rate $\dot{m}_{slide}$ and through 2) any leaks around the edges of the LEU with temperature T_b and mass flow rate $\dot{m}_{leak}$. Similarly for region *a*, warm ambient air enters the control volume with temperature T_{amb} and mass flow rate $\dot{m}_{in}$, and cool air enters the control volume with temperature T_b and mass flow rate $\dot{m}_{slide}$. Air leaves the control volume and enters the evaporator with temperature T_a and mass flow rate $\dot{m}_{fan}$. The evaporator fan drives the air mass flow throughout the system with a constant flow rate, $\dot{m}_{fan}$. The individual air mass flow rates are based on the pressures in regions *a* and *b* as shown in Equation (1) where P_a and P_b are the pressures of the air inside regions *a* and *b*, respectively, and R_{in} , R_{leak} , R_{slide} and are the hydraulic resistances of the inlet, leaks around the LEU edges, and the slide, respectively.

$$\dot{m}_{in} = -\frac{P_a}{R_{in}}, \quad \dot{m}_{leak} = \frac{P_b}{R_{leak}}, \quad \dot{m}_{slide} = \frac{P_b - P_a}{R_{slide}} \quad (1)$$

The fan on the evaporator is assumed to provide a constant mass flow rate of air. The resistance of the slide, R_{slide} , as a function of its percentage opening, shown in Equation (2), is most critical to the dynamics of the LEU.

$$R_{slide} = \frac{\rho}{2C_d^2 A_0^2} \dot{m}_{slide} \quad (2)$$

C_d is the drag coefficient of the slide, ρ is the density of the air, and A_0 is the area of the flow opening. The drag coefficient is calculated as a function of slide opening in Equation (3),

$$C_d = \frac{1.1 + 0.02 \left(\frac{W}{H} + \frac{H}{W} \right)}{\sqrt{1 - (1 - \beta)^4}}, \quad \beta = \frac{A_0}{A}, \quad (3)$$

where W is the width of the opening, H is the height of the slide, and β is the normalized slide opening. The pressures of the air in regions *a* and *b* are determined based on an electrical analogy using Equations (4) and (5) where C_a and C_b are the hydraulic capacitances of the regions.

$$\frac{dP_a}{dt} = \frac{1}{C_a} \left[-\dot{m}_{fan} - \frac{P_a}{R_{in}} + \frac{P_b - P_a}{R_{slide}} \right] \quad (4)$$

$$\frac{dP_b}{dt} = \frac{1}{C_b} \left[\dot{m}_{fan} - \frac{P_b}{R_{leak}} - \frac{P_b - P_a}{R_{slide}} \right] \quad (5)$$

The same control volume approach was used to determine the average air temperatures in each region. Equations (6) and (7) were derived from an energy balance for each control volume.

$$\frac{dT_a}{dt} = \frac{1}{\rho V_a c_p} \left[-\frac{P_a}{R_{in}} T_{amb} + \frac{P_b - P_a}{R_{slide}} T_b - \dot{m}_{fan} T_a \right] \quad (6)$$

$$\frac{dT_b}{dt} = \frac{1}{\rho V_b c_p} \left[\dot{m}_{fan} T_{low} - \frac{P_b - P_a}{R_{slide}} T_b - \frac{P_b}{R_{leak}} T_b \right] \quad (7)$$

While these dynamics are highly nonlinear, the new design of the LEU was intended to provide nearly linear dynamics between return air temperature and slide opening over a range of slide positions. Equation (8) shows the desired first order dynamics between the return air temperature, T_a , and the slide opening, β ,

$$\tau \frac{dT_a}{dt} + T_a = K\beta \quad (8)$$

where τ is the time constants and K is the gain. Equation (6) can be rearranged as shown below

$$\tau \frac{dT_a}{dt} + T_a = -\frac{P_a}{\dot{m}_{fan} R_{in}} T_{amb} + \frac{P_b - P_a}{\dot{m}_{fan} R_{slide}} T_b \quad (9)$$

where $\tau = \rho V_a c_p / \dot{m}_{fan}$. The right hand side of Equation (9) can be simplified using Equation (1) as shown in Equation (10).

$$\tau \frac{dT_a}{dt} + T_a = \frac{\dot{m}_{in} T_{amb} + \dot{m}_{slide} T_b}{\dot{m}_{fan}} \quad (10)$$

Assuming that the hydraulic dynamics are much faster than the thermal dynamics, resulting in $\dot{m}_{in} = \dot{m}_{fan} - \dot{m}_{slide}$, Equation (10) can be further simplified to Equation (11) where $\dot{m}_{slide}$ is a function of β .

$$\tau \frac{dT_a}{dt} + T_a = T_{amb} + \frac{T_b - T_{amb}}{\dot{m}_{fan}} \dot{m}_{slide}(\beta) \quad (11)$$

The ambient temperature, T_{amb} , is a constant offset and can be ignored for the current analysis. The temperature of region b , T_b , is assumed to be constant, which is verified with the experimental data below. Therefore, the right hand side of Equation (11) can be expressed as a linear function of slide opening with the form $K\beta$ if the air mass flow rate through the slide is a linear function of β . The experimental data below shows that this linear approximation of the LEU dynamics is valid for a specific range of slide openings. This linearization was not possible with the previous LEU design due to the complex flow dynamics caused by the physical design of the system. The previous design had tight turns and sharp corners that created turbulence and vortices in the air flow, resulting in non-monotonic and nonlinear dynamics over the entire range of actuator positions (Otten *et al.* 2010).

The current LEU design, which uses a linear slide as an actuator, has streamlined flow paths which result in monotonic and nearly linear behavior over a large range of actuator positions. Figure 4 shows the time responses of the return and supply air temperatures as the slide is opened in 10% increments. It is clear from this figure that the LEU dynamics are monotonic but nonlinear; the return air temperature plateaus at a slide position of 30% open and only the supply air temperature continues to decrease for larger slide openings. Figure 5 shows a closer look at the 0 – 25% open region with the slide opened in increments of 2.5%. Over this actuation range, the supply air temperature is fairly constant, justifying the assumption $T_b = \text{constant}$, and the return air temperature decreases approximately linearly. Figure 5 also shows that an increase in slide opening from 0 – 2.5% does not significantly affect the return air temperature. A slide position of 2.5% is actually fully closed and 0% creates an overlap between the slide and the boundary wall; therefore 2.5% open is the lower limit in slide position.

Since the 2.5 – 30% open range of slide positions nearly spans the achievable return air temperature range and is approximately linear, only this range will be used for temperature control. Figure 6 shows a linear fit to the measured data represented by Equation (12).

$$T_{return} = -0.3\beta + 23 \quad (12)$$

The transient response from Figure 5 provides the time constant for the first order approximation of the LEU dynamics from Equation (8). The average time constant was determined to be 10 seconds by evaluating the time response to the 2.5% step in slide opening for each step in Figure 5. It is important to note that the slide itself is actuated very quickly with dynamics that are approximately two orders of magnitude faster than the dynamics of the return air temperature. The gain, K , in Equation (8) is the slope of the linear approximation from Equation (12) which is -0.3°C per % open. Therefore, the plant model for the dynamics of the LEU can be approximated as a linear first order differential equation for the 2.5 – 30% open range of slide positions as shown in Equation (13).

$$10 \frac{dT_a}{dt} + T_a = -0.3\beta \quad (13)$$

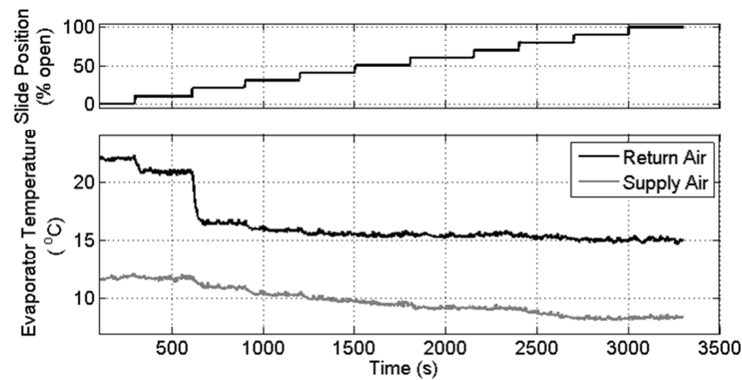


Figure 4: Evaporator return temperature as a function of time for 10% step increases in slide opening.

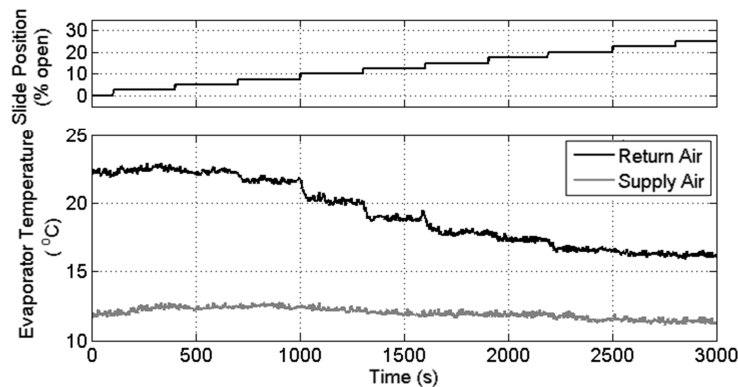


Figure 5: Closer look at evaporator return temperature with 2.5% step increases in slide opening.

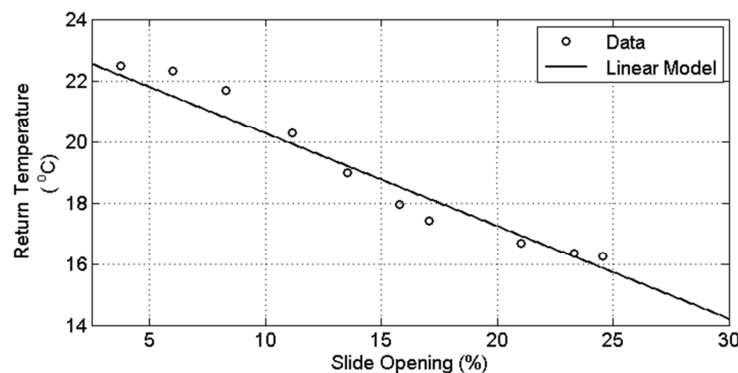


Figure 6: Evaporator return air temperature as a function of slide opening.

3. VIRTUAL CONTAINER MODEL

As discussed in the introduction, the LEU connects the virtual container model (VCM) with the physical AC&R system that is being tested. The VCM simulates the environment in which the AC&R system would interface in a specific application. The VCM is run in real-time to simulate how the experimental system *would* cool the virtual container. Figure 7 shows the inputs and outputs to the VCM, which in this case is a refrigerated transport container.

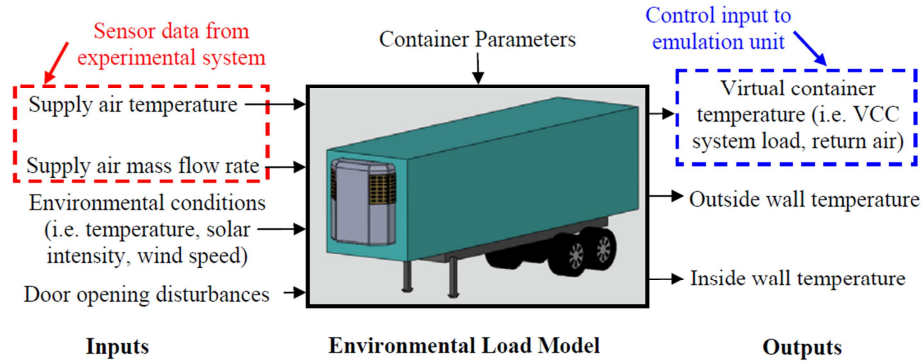


Figure 7: Inputs and outputs of virtual container model.

The inputs to the VCM include measured supply air temperature and air mass flow rate data from the experimental test system as well as environmental conditions such as ambient temperature, solar radiation, wind speed, and door openings which are specified based on anticipated real world conditions. The VCM outputs the internal air temperature of the container which is sent to the LEU as well as the internal and external wall temperatures of the container. The VCM assumes a uniform interior air temperature, uniform wall surface temperatures, and one-dimensional heat conduction through the walls. The set of first order differential equations used to determine the interior air temperature and wall temperatures are given in Equations (14) – (16) where the AC&R system capacity, $\dot{Q}_{vcc}$, the door opening load, $\dot{Q}_{door}$, and the solar load, $\dot{Q}_{solar}$, (where α is absorptivity and I_{solar} is solar intensity) are calculated using Equations (17) – (19).

$$\frac{dT_{space}}{dt} = \frac{\dot{Q}_{conv,in} + \dot{Q}_{door} - \dot{Q}_{vcc}}{(\rho V c_p)_{air}} \quad (14)$$

$$\frac{dT_{wall,in}}{dt} = \frac{\dot{Q}_{cond} - \dot{Q}_{conv,in}}{(\rho V c_p)_{wall}} \quad (15)$$

$$\frac{dT_{wall,out}}{dt} = \frac{\dot{Q}_{solar} + \dot{Q}_{conv,out} - \dot{Q}_{cond}}{(\rho V c_p)_{wall}} \quad (16)$$

$$\dot{Q}_{vcc} = \dot{m}_{air} c_p (T_{space} - T_{supply}) \quad (17)$$

$$\dot{Q}_{door} = f(T_{amb}, T_{space}, A_{door}) \quad (18)$$

$$\dot{Q}_{solar} = \alpha I_{solar} \quad (19)$$

Additional information on the calculation of these thermal loads, including $\dot{Q}_{conv,in}$, $\dot{Q}_{conv,out}$, and $\dot{Q}_{cond}$, can be found in Li and Alleyne (2009), and the validation of this model can be found in Otten *et al.* (2010).

4. CONTROL DESIGN

Figure 8 shows a high level block diagram of the control architecture for each LEU. The VCM provides the desired return air temperature based on the measured air mass flow rate and supply air temperature from the experimental test system. The LEU controller determines the slide position needed to provide the desired return air temperature using the measured return air temperature as its feedback signal. The slide controller moves the slide to track the desired slide position using the measured slide position as its feedback signal.

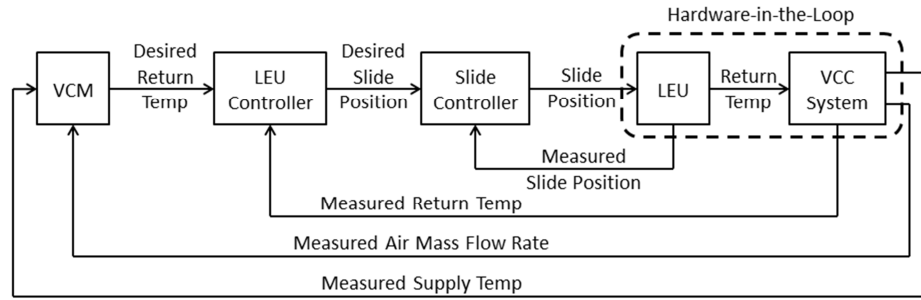


Figure 8: Schematic of high level control structure.

The evaporator return air temperature controller for the previous LEU design required feedforward control and gain scheduling in order to compensate for the highly nonlinear and non-monotonic open loop behavior of the physical system (Otten *et al.* 2010). As described in Section 2, the current LEU design was chosen to provide more linear and monotonic open loop behavior, which was achieved in the 2.5 – 30% slide opening range, resulting in the first order dynamic shown in Equation (13). Since the open loop behavior of each LEU is nearly linear in this operating range, a proportional-integral-derivative (PID) controller is sufficient in controlling the evaporator return air temperature. If the desired return air temperature is outside of the range of achievable temperatures, the slide will saturate at 2.5% or 30% open; therefore linear feedback anti-windup was added to prevent integral windup (Levine 2011). The closed loop Ziegler Nichols tuning method (Ogata 2010) was used to determine the gains for the PID controller, shown in Equation (20),

$$K_p = 0.6K_{cr}, \quad K_i = 2 \frac{K_p}{P_{cr}}, \quad K_d = \frac{K_p P_{cr}}{8} \quad (20)$$

where K_{cr} is the critical gain at which the output signal (return air temperature) oscillates with a constant amplitude and P_{cr} is the period of the oscillations. The critical gain was determined to be 7 with a period of 15 seconds. The anti-windup gain, K_s , was chosen to be the same as the integral gain. After slightly adjusting the gains to improve performance, the PID controller, shown as a Laplace transfer function in Equation (21), was used for both LEUs in the ME-LEU.

$$G_c(s) = 4 + \frac{0.3}{s} + 7.5s \quad (21)$$

The slide is controlled using a similar PID controller and is actuated using a DC motor and a rack and pinion system as seen in Figure 2. A digital optical encoder is used to measure the actual position of the slide which is used as the feedback signal in the controller.

Figure 9 shows the ability of a single LEU to track a desired return air temperature by adjusting the position of the slide. In this test, the desired return air temperature was specified rather than being determined by the VCM. When the desired return air temperature is within the achievable temperature range (shown in Figure 9 for $t=250$ until $t=1100$ seconds), the return air temperature response is fast, with a time constant of approximately 5 seconds and minimal overshoot. Due to the slight nonlinearity in the open loop LEU dynamics, as the desired temperature reaches the limits of the achievable range, the temperature response is slower. If the desired temperature is outside the achievable range (see $t=1100$ until $t=1300$ seconds in Figure 9), the slide saturates at either 2.5% or 30% open and waits for the desired temperature to reenter the achievable range. Due to the anti-windup feature in the return air temperature controller, once the desired temperature re-enters the range of achievable temperatures, the LEU is able to quickly track the desired temperature with minimal overshoot. With a time constant of 5 seconds, the LEU is able to accurately track temperature changes of air-filled rooms larger than about 4 m^3 assuming an air density of 1.2 kg m^{-3} and a specific heat of 1 kJ(kg K)^{-1} .

5. CASE STUDIES

The following case studies demonstrate the performance of each individual LEU and the added operational capabilities that the ME-LEU provides to a dual evaporator test system. Figure 1 shows the ME-LEU mounted to the evaporators of a vapor compression cycle (VCC) test system at the University of Illinois at Urbana-Champaign.

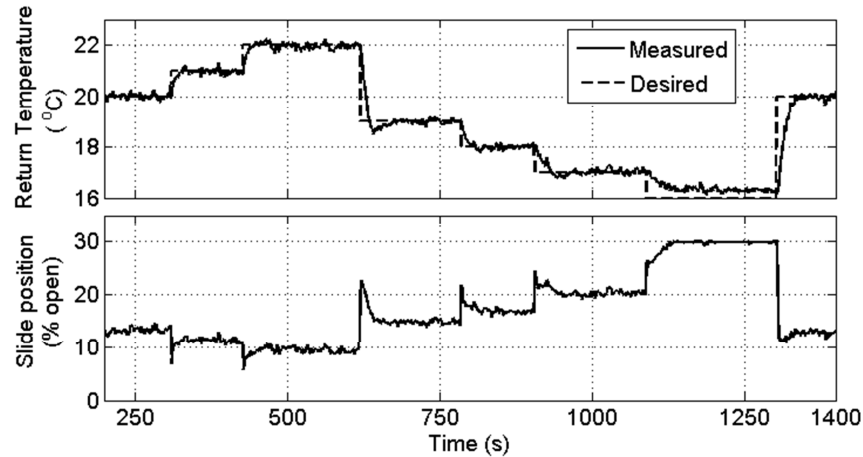


Figure 9: Reference temperature tracking and slide position dynamics.

The system has a maximum cooling capacity of 1kW which restricts the dynamic range of achievable evaporator return air temperatures to approximately 6-7°C in the ME-LEU. On larger systems, where the difference between the evaporator air return and supply temperatures is greater, larger temperature ranges are achievable. The air mass flow rate was kept constant for each case study. While the following case studies were performed using open loop control of the refrigerant dynamics, the ME-LEU provides the ability to test more advanced control algorithms with different loads applied to each evaporator, which is a considerably more difficult control problem.

5.1 Dual LEU Capabilities and Refrigerant Effects

While LEUs are certainly useful for single evaporator studies, one of the primary advantages of this ME-LEU is its ability to provide different loads to each evaporator in a dual-evaporator system, despite having the entire system in a single lab space. Region 1 in Figure 10 shows the ability of the ME-LEU to provide a 5°C difference between the return air temperatures to each of the evaporators.

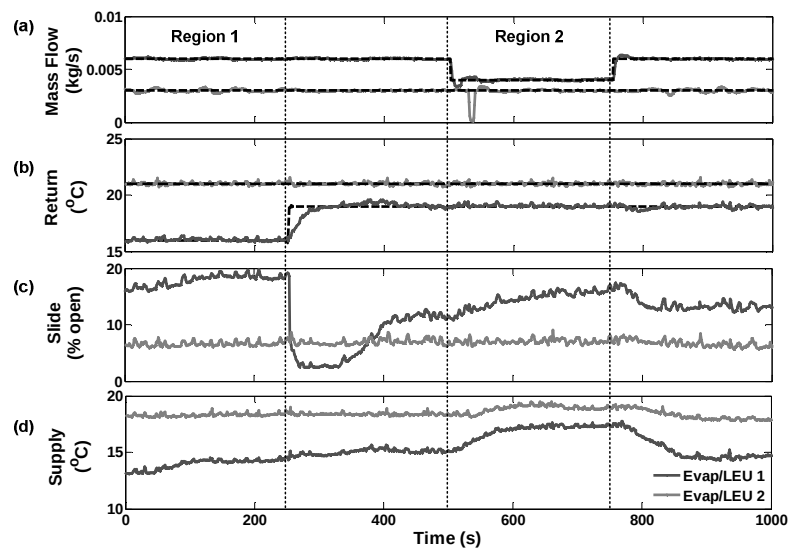


Figure 10: a) Refrigerant mass flow rate through each evaporator. b) Return air temperature to each evaporator. c) Slide opening for each LEU. d) Supply air temperature from each evaporator. Desired reference signals shown with a dashed line in (a) and (b).

Due to the near linear dynamics of the LEUs and the use of PID control, the LEUs easily compensate for changes in refrigerant side operating conditions while maintaining the desired return air temperature (Figure 10b). Region 2 in Figure 10 shows that when the refrigerant mass flow rate to evaporator 1 is reduced from 6 g/s to 4 g/s (Figure 10a), resulting in an increase in supply air temperature, the LEU simply increases the slide opening (Figure 10c) to

maintain the desired return air temperature. Refrigerant side conditions do, however, influence the range of achievable return air temperatures. In Region 2, because the supply air temperature from the evaporator increases as the refrigerant mass flow rate is reduced, the lower bound of achievable return air temperature increases. Therefore, the refrigerant side operating conditions should be considered when determining the range of achievable return air temperatures. It is also important to note that the ambient temperature of the laboratory affects the upper bound of the achievable return air temperatures. Heaters may be added to the inlet of the LEU to raise the upper bound but have not been used in this study.

5.2 Environmental Load Simulation and Emulation through Hardware Actuation

In the previous case study, the desired temperature was a user defined value. However, the benefits of the ME-LEU are greatly extended by coupling the ME-LEU with the VCM discussed in Section 3. Figure 11 shows the results of a study in which two similar spaces were simulated using two VCMs where space 1 had a larger thermal load than space 2. The figure shows the open loop cooling of the two spaces from an initial temperature of 21°C. At $t=1500$, $t=2500$, and $t=3500$ seconds, a 100 second door opening is simulated, resulting in a sudden rise in the temperature of the spaces. The LEUs are able to supply air at the temperature determined by the VCMs, thereby effectively providing the thermal loading of the cooled spaces to the test system. While this case study simulates two similar rooms with different thermal loads, the VCMs and the ME-LEU provide the ability to model many types of spaces with highly varying dynamics, thermal loads, and disturbances. Prior to the addition of the ME-LEU, the condenser and both evaporators on the VCC test system interfaced with a shared temperature, which is unrealistic for most VCC systems. The ME-LEU adds the capability to provide independent temperatures to the heat exchangers which significantly extends the set of real world systems that can be studied using the experimental VCC test system. In the future, advanced control techniques will be developed and tested on this test system with the added ability to provide independent loading of each evaporator.

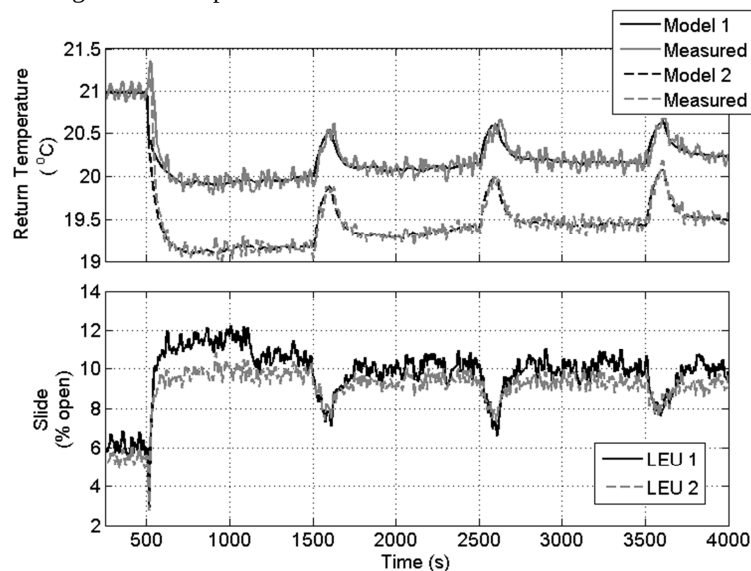


Figure 11: Return temperature tracking using dual virtual container models.

6. CONCLUSIONS

Load emulation units are an effective way of providing variable thermal loads to air conditioning and refrigeration systems while minimizing the need for large, costly environmental testing chambers. The virtual container model simulates the environmental thermal loads and the dynamics of the cooled space, determining the desired return air temperature which is tracked through hardware actuation of a slide in the LEU. The application of a dual LEU extends the capabilities of a dual-evaporator vapor compression test system by permitting independent loading of each evaporator. This provides the ability to test additional operating conditions for the development of advanced control techniques for multi-evaporator air conditioning and refrigeration systems.

Future work will introduce thermal coupling between spaces by physically connecting two LEUs with variable insulation and will use the ME-LEU to test how refrigerant-side multi-evaporator control algorithms respond to

environmental disturbances and varying loads which affect each evaporator in the system. One of the primary advantages of using advanced control algorithms is to better respond to these disturbances which enter via the environment or container that is being cooled. Although various control schemes have been proposed for dual and triple-evaporator VCC systems (Lin and Yeh 2009) (Wu, Zhou and Shiming 2005) (Elliott and Rasmussen 2009), the performance of these controllers is typically demonstrated for a specific environmental chamber or container in a laboratory setting. With the ME-LEU, we have the ability to introduce a wider range of disturbances as well as consider spaces of many sizes and thermal characteristics to more rigorously test the performance and robustness of various ME-AC&R control schemes.

NOMENCLATURE

A	area	(m^2)	Subscripts	
C_d	drag coefficient	$(-)$	amb	ambient
c_p	specific heat	$(kJ (kg K)^{-1})$	d	derivative
K	gain	$(-)$	i	integral
$\dot{m}$	mass flow rate	$(kg s^{-1})$	p	proportional
$\dot{Q}$	heat transfer rate	(kW)		
T	temperature	(K)		
t	time	(s)		
V	volume	(m^3)		
β	normalized slide opening	$(-)$		
ρ	density	$(kg m^{-3})$		
τ	time constant	(s)		

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